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Freeman Neurodynamics

The Past 25 Years

Abstract: *Walter Freeman established a unique approach for interpreting brain processes, perception, cognition, and intentionality. Freeman's neurodynamics approach evokes the concepts of mass action and synchrony in neural populations, and even today is far ahead of the field of dynamical systems in hierarchical brain models. He summarized the essence of his views on the physiology of perception in a landmark paper on the pages of Scientific American in 1991. He spelled out the main components of his neurodynamics theory in that essay, which became a classic in brain theory and cognition. His approach has been hailed by many and attracted a large number of scientists all over the world. At the same time there*

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were fellow scientists who objected to Freeman's approach and dismissed its basic tenets. This love or hate relationship followed Freeman until his death in his home in Berkeley, on 24 April 2016. In the present contribution we review the progress of Freeman's theory of neurodynamics over the past 25 years. We describe his work in light of new developments in experimental and theoretical approaches to brain dynamics, establishing new directions in computational neuroscience, cognitive monitoring, computational and mathematical modelling, field theories of cognition and intelligence, and his quest towards inventing novel engineering applications.

1. Walter Freeman's Legacy: A Summary

Walter Freeman was a giant in the field of neuroscience, and his visionary work significantly shaped the field over the past 60 years. He had a profound impact on neuroscience all over the world. 'Freeman neurodynamics' is a concept that is widely known and respected among neuroscience researchers everywhere. Many of his revolutionary ideas in computational neuroscience were ahead of their time by decades. His main quest was to decipher the operation of brains. In his landmark paper in *Scientific American* (Freeman, 1991), Freeman summarized his main findings as follows: 'The brain transforms sensory messages into conscious perceptions almost instantly. Chaotic, collective activity involving millions of neurons seems essential for such rapid recognition.' He made crucial advances in several key areas:

A. Physiology of sensory processing in mammals, with a special emphasis on olfaction.
B. Higher-level cognitive functions, cognition, large-scale brain systems/networks.
C. Hierarchical modelling of the action–perception cycle; known widely as Freeman K-sets.
D. Singular transitions in the cortex modelled as coherence–decoherence switches described by tools of quantum field theory (QFT).
E. Broader issues in systems theory, intentionality, and the unity of the mind-brain-body.

In this review paper we describe the development of all these areas since the publication of Freeman's landmark work 25 years ago. Many of his visionary predictions have been confirmed since then, and there are many open questions ahead.

2. Freeman Neurodynamics

2.1. Mesoscopic Brain Dynamics

Walter Freeman's crucial contributions in neuroscience lie in the analysis of oscillations in coherent activity of large populations of neurons, known as the local field potential (LFP) when measured within the brain, and the electroencephalogram or electrocorticogram (EEG or ECoG) when measured on the surface of the scalp or brain, respectively. This activity is referred to as mesoscopic (middle-sized view) because neuronal populations, at a level midway between single neurons and whole brain areas, generate it. Freeman discovered that the study of mesoscopic activity can inform us about the workings of the underlying neural tissue and that these emergent population-level phenomena are involved in perception in all vertebrates.

At the foundation of Freeman's theory of *mass action* is the idea that the interconnectedness of large populations of neurons leads to population-level effects that can support large-scale dynamical states. Freeman began his journey into these ideas with his first publication as a young assistant professor at UC Berkeley in 1959. He offered the first description of a cortical oscillation distributed across a brain region in the prepyriform cortex, often referred to as the primary olfactory cortex. This work laid the foundation for his future studies in which he described oscillatory responses to electrical stimulation in various cortical areas. He combined these recordings of the local field potential with analysis of single neuron firing patterns, which led him to characterize the relationship between action potentials and the collective or emergent properties of the local field. This relationship is described by an S-shaped function, called an asymmetric sigmoid non-linearity, which forms the foundation of Freeman's theory. The functional form of the sigmoid is derived from the probability of pulse firing conditional on ECoG amplitude; see Figure 1a.

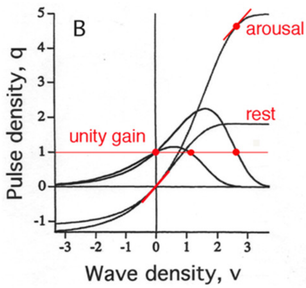

Fig. 1a.

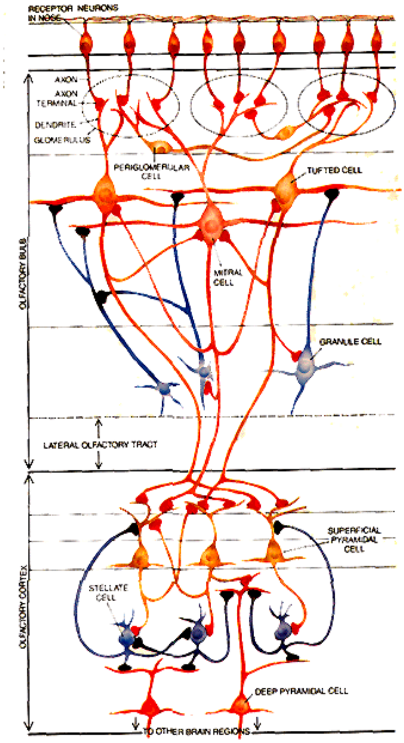

Fig. 1b.

Figure 1. a. The sigmoid curve describing the relation of pulse density (q) to wave density (v); the functional form of the sigmoid is derived from the probability of pulse firing conditional on ECoG amplitude and given as $q(v) = Qm\ (1 - \exp[(1 - ev)/Qm])$. The value of the upper asymptote depends on the level of behavioural arousal: here Qm = 1.8 for the resting state and 5 for the arousal state. Calculation of the gain (dq/dv) gave two steady state operating points, one at zero normalized wave amplitude and the other at a positive wave density, which gave the level of the background activity (from Freeman, 1979). b. Freeman's classical result concerns the characterization of the olfactory system as a multilayer network of neurons, starting from the receptor neurons in the nose, projecting to the olfactory bulb, with the signals produced there being sent to the olfactory cortex via the lateral olfactory tract. The olfactory cortex is connected to other brain regions, which together contribute to the formation of multisensory percepts. In this schematic diagram, red shading signifies that a neuron excites another cell, black shading shows that a neuron inhibits another (from Freeman, 1991).

At the same time, Freeman was hard at work building some of the first computational models of cortical networks. He was influenced by the work of Norbert Wiener and Aharon Katchalsky, but also by Leon Chua, Morris Hirsch, and Lotfi Zadeh, his colleagues at UC Berkeley. These models were based on the responses of neural masses in the olfactory bulb and pyriform cortex to a small electrical stimulus to the olfactory tract or to the olfactory nerve, which delivers activity from the olfactory sensory epithelium in the nose to the olfactory bulb, and the olfactory cortex, as illustrated in Figure 1. The mutual activity of interacting populations of neurons produces spatio-temporal oscillations in the olfactory bulb and the cortex, which manifest the meaning of the sensory stimuli for the subject.

A series of experiments led Freeman to conclude that mutual excitation between neurons serves to stabilize the system rather than drive it into instability via runaway excitation as had been assumed. How did the stability work? The key was the refractory period of individual neurons. When neurons fire an action potential to send messages down axons to other neurons and other brain regions, the ~1 ms long pulse is followed by a brief period (~1 ms) in which the neuron cannot fire another action potential and an additional few ms which requires larger excitation to fire an action potential. This property in a population of neurons prevents the runaway excitation predicted by feedback excitation onto excitatory neurons.

By the late 1970s, sampling rate increases and cost decreases for data acquisition boards and amplifiers meant that Freeman was able to expand his recording scope to examine the activity at up to 64 sites within a cortical structure simultaneously, which he called amplitude modulation (AM) patterns; see Figure 2. This simple advance led to a revolution in his thinking about how brains interpret the sensory world. He began these experiments to test the hypothesis, popular in the olfactory research community, that the information from the receptor sheet in the nose should be translated to reliable chemotopic representations of odours in the olfactory bulb. The experiments showed that the EEG waveform was very similar across all of the electrodes in a cortical area. Volume conduction as a source of this similarity could be ruled out because the phase of the common ‘gamma’ oscillation (40–100 Hz) varied across the electrodes.

The amplitude modulated (AM) spatial pattern across the surface of a cortical area was quasi-stable, but it was not an odour representation in the sense that repeated stimuli from the environment produce a repeated pattern of activation on a sensory sheet. Rather, the patterns depended on associations made with an odour stimulus, and the patterns changed when associations were changed or new associations were made, even those that did not involve that odour. Patterns also drifted over time, consistent with the ways in which life experience changes the meaning of old knowledge. Further analysis revealed that these patterns could emerge without reinforcement when the odour stimulus was paired with delivery of a neurochemical (norepinephrine) known to be involved in learning. Finally, these spatial AM patterns showed a point of origin of the phase of the common waveform on each sniff. In the olfactory bulb, this point varied randomly from sniff to sniff and showed no discernible pattern or sequence across trials. Similar phenomena were later shown in visual, auditory, and somatosensory cortices.

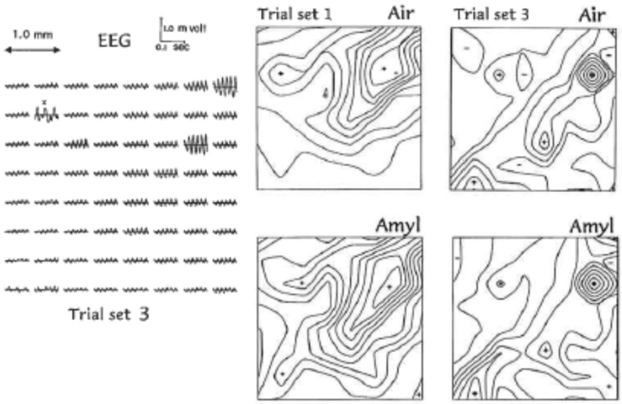

Figure 2. Illustration of amplitude modulation, which Freeman called amplitude modulation (AM) patterns, of the filtered ECoG signals across the gamma band for an array of 8 x 8 electrodes (3.5 x 3.5 mm) on the olfactory bulb. The insert on the left shows the ECoG oscillations, while the frames on the right show the AM patterns for air and odorant (amyl acetate). There is a clear change in the AM patterns with the reinforced conditioned stimulus. The horizontal comparisons, i.e. Trial set 1 vs. Trial set 3, show the changes that occur over time as the rabbit gains experience and learns new odours (from Freeman and Schneider, 1982).

It is worth noting that Freeman's discovery of these dynamic meaning-associated spatial AM patterns preceded discovery of the odour receptor genes. Following that discovery, anatomical and imaging work showed that a relatively stereotyped input pattern is delivered to the olfactory bulb by sensory receptor neurons, and that these input patterns are indeed chemotopic odour representations. His work showed that these patterns are quickly converted to meaning-related activity, and later work from his lab confirmed these types of effects in other sensory systems in rabbits and cats.

The mid- to late-1980s was witness to an explosion in understanding of nonlinear dynamical systems, and work by Prigogine and Haken heavily influenced Freeman's thinking in this field. The presence of dynamical stability driven by excitation as exemplified in the sigmoid relationship between spiking and wave activity, emergent meaning-dependent spatial patterns, and the random source of origin within the olfactory bulb all led Freeman to propose that this system acted like chaotic systems found in nature. Key to these ideas from a physiological perspective is that cortical activity is aperiodic in nature. His work supported the idea that brain regions do not hierarchically process information but rather work together as an emergent system that is influenced only a small amount by the sensory stimulus and a much larger amount by ongoing neural states. Also, in order for the olfactory bulb to operate as a meaning-dependent structure, it uses 'preafferent' input from higher-order areas to prepare itself for an expected stimulus. All of Freeman's more recent studies continue to provide a foundation for deep and insightful enquiry into the neurodynamical mechanisms that support cognitive functions in animals and humans. (See also the papers by Kay, and Mannino and Bressler, in this issue.)

2.2. Higher-Level Cognitive Dynamics — Intentional Neurodynamics and the Cinematographic Theory of Perception

From the 1990s to the present, Freeman showed that the 'cortical code' consists of repetitive frames of spatial amplitude modulation (AM) patterns progressing in a meaningful sequence. His findings during this time led him to postulate the intentional action–perception (A–P) cycle, which starts with the subject creating an hypothesis about the external world, in the context of its internal goals and desires. Based on this hypothesis, the subject makes an action, which is the intentional act of reaching out into the environment. As the result of the action and monitoring of the corresponding sensory input, the hypothesis is either confirmed or rejected. In both cases the resulting percepts are incorporated in the internal 'worldview' of the subject, and the internal world model is updated. The intentional A–P cycle is then ready to start again with a new set of actions.

Freeman considered the brain not primarily as an information processing organ, but as a system for creating meaning, as represented in spatial amplitude modulated patterns of brain activity. Each such pattern manifests a meaning to the animal, embodying immediate goals, such as finding food, shelter, or a mate, the achievement of which requires acquisition of information from the environment. Meaning implies context, but also intention, which is the creation and projection by the brain of alternative future states, desired or feared. Hypotheses about the future are constructed in attractor dynamics by extrapolation from past experience, and they serve to control choices and directions of actions in the present. They are tested by actions into the environment and are evaluated and updated by the sensory consequences of the actions. The operations of predicting, planning, acting, detecting, and learning comprise the process of intentionality.

Freeman considered the neurodynamics of intentionality to occur at various levels in the evolution of brains. He considered amphibians, among vertebrates, as the animals at the lowest evolutionary level that have intentionality to support their movement and navigation in the environment. Figure 3 shows the schematics of the amphibian brain with olfaction dominating sensory processing, the hippocampus for space-time associative learning, and the striatum connecting to the pyriform cortex for motor actions. Mammalian intentionality is assisted by the limbic system as outlined in Figure 4. The interaction of the body and the environment is shown, on the one hand, as the body reaching out to the environment through motor action, and observing the results of the action using sensory receptor modalities, on the other hand. The sensory systems include not only external but also internal senses through the proprioceptive loop. The central role of the entorhinal cortex is shown, coordinating the interaction with the hippocampus and sensory systems, and controlling the motor system.

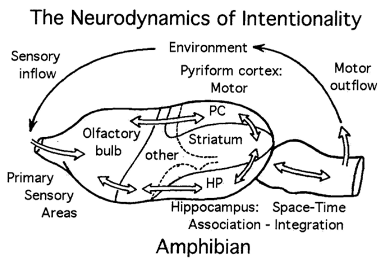

Figure 3. The neurodynamics of intentionality for an amphibian (adopted from Freeman, 1999).

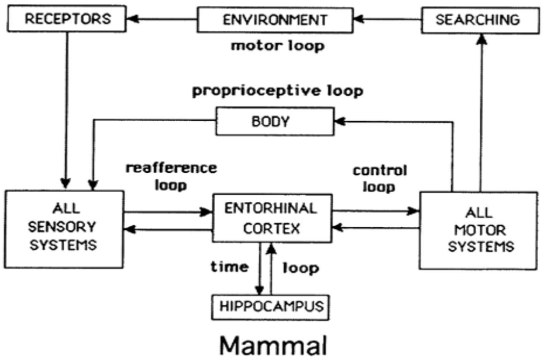

Figure 4. The neurodynamics of intentionality for a mammal (adopted from Freeman, 1999).

Freeman's understanding of intentionality derived, he said, from a need to 'fill the explanatory gap between my electrophysiological data and the goal-directed behavior of animals' (Freeman, 2008). A key aspect of intentional behaviour is that an action is preceded by a goal that shapes it, and the perception of a stimulus is preceded by attentive awareness that prefigures it. Freeman concluded that only a small fraction of intentional behaviour is conscious, and 'that fraction is much too small a sample to be trusted with the management of intent and assimilation' (Freeman, 2007). He commented that, luckily, we don't have to rely on our consciousness to make decisions, otherwise we would be paralysed. We make these decisions in a fraction of a second, and Freeman argued that the chaotic nature of cortical processing makes this possible. Chaotic systems allow large populations of oscillating neurons to switch modes almost instantaneously. (A further discussion on Freeman's views on intentionality is given by Liljenström in this issue.)

His work over the past 25 years led to detailed characterization and quantification of rapid changes in collective spatio-temporal brain dynamics. Applying Hilbert transform-based analytic techniques to ECoG and EEG data, he observed extended periods of large-scale synchrony in cortical oscillatory activity, despite delays imposed in communication by propagating action potentials. Extended periods of large-scale synchrony in an entire brain hemisphere are interrupted by brief episodes of desynchronization, when the analytic amplitude diminishes and rapidly propagating phase gradients, phase cones, are formed over large cortical areas. The macroscopic phase cones are initiated at a given point of time and space on the cortex. These initiation points can be viewed as intermittent singularities when the power of the signals diminishes and the activity becomes desynchronized. The singularities initiate a rapid change in the state of the cortex, manifested in rapid increase in the analytic power and in the formation of AM patterns, which are the manifestations of large-scale order and synchrony.

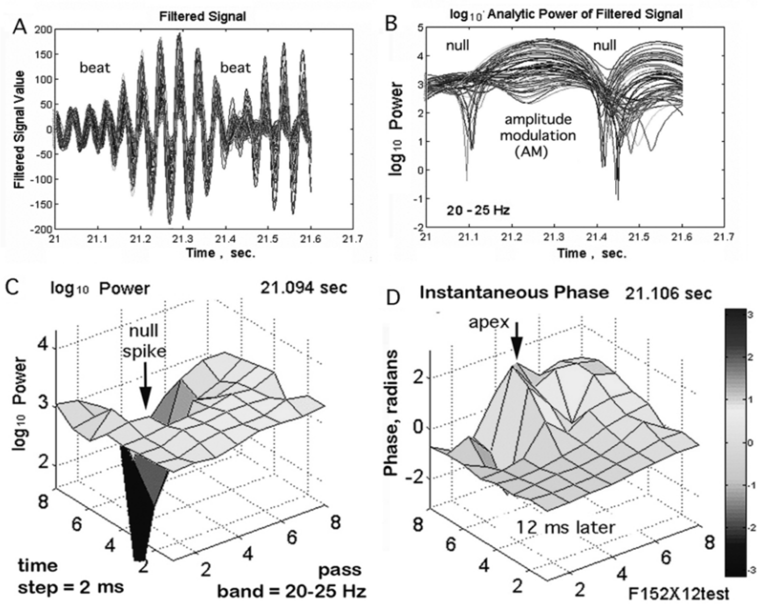

Figure 5. Structure of the ECoG analytic amplitude and phase determined using Hilbert transform for the signals measured over the rabbit visual cortex and filtered in the 20–25 Hz band. A. Overlay of the 64 band-passed ECoG signals showing beating patterns at about 300 ms intervals. B. Display of the log analytic amplitudes of the 64 ECoG channels; the amplitude of a number of channels drastically decreases at time instances corresponding to the beats. C. 3D view of a null spike when the log power drastically diminishes. D. Illustration of a phase cone showing its apex near the site of the null spike (that occurred 12 ms earlier) (adapted from Freeman, 2006).

These rapid changes in cortical activity led Freeman to the cinematographic theory of perception, which proposes that during perception the cortex enters a series of brief quasi-static states, like movie frames, and the momentary desynchronization acts like a shutter separating one frame from the other. Spatial AM patterns last for ~150–200 ms (repeating at a rate in the theta frequency range). Each spatial pattern is then destabilized and collapses for a period of ~10–20 ms, after which a new AM pattern emerges, corresponding to new (possibly altered) external/internal conditions. Thus, ordered activity patterns are separated by sequenced lapses, where the lapses serve as rapid and repeated transitions to new patterns. Figure 5 illustrates various aspects of the synchronization–desynchronization sequence using results of ECoG measurements from an 8 x 8 array of intracranial electrodes.

2.3. Mathematical and Computational Modelling — Freeman K-Sets Implemented in ODE and Graph Theory Domains

One of the key features of Freeman's discoveries lies in the mathematical and computational models utilized in the analysis of brain dynamics. His 1975 book, *Mass Action in the Nervous System*, predated the interest in chaos theory to explain brain function. It is still today the best example of a successful integration of circuit theory mathematics with the physiology of neural assemblies. Freeman has modelled brain dynamics in a long list of seminal papers spanning over 40 years, using arrays of nonlinear oscillators now called Freeman Katchalsky/K-sets. Freeman K-sets are examples of population models with microscopic, mesoscopic, and macroscopic levels of granulation.

K-sets have been analysed, extended, and implemented in various fields in the past decades. Freeman's breakthrough contribution of K-sets lies in a compelling characterization of the relationship between neural structure, spatio-temporal neurodynamics, neural functions, and cognitive behaviours. Freeman K-sets are still unparalleled in the way they provide a hierarchy of models linking microscopic, mesoscopic, and macroscopic structures and behaviours.

Fundamental structures and dynamics of basic Freeman K-sets are illustrated in Figure 6. The existence of neural populations is manifested in the micro-column structure of cortical tissues. The basic unit of neural populations in K-sets includes several 1,000s of neurons, representing the mesoscopic level, which corresponds to the interaction of neural populations with convergent dynamics. These populations can be either excitatory or inhibitory (KI). Coupled excitatory and inhibitory populations exhibit narrow-band oscillators due to negative feedback (KII). The interaction of several such oscillatory units with different dominant frequencies (KIII) can produce non-convergent (chaotic) oscillations. KIII sets play an important role in the interpretation of amplitude modulation (AM) patterns of neural activity as described previously in Figure 2. Accordingly, negative feedback in neural populations creates the narrow-band (e.g. beta or gamma band) carrier wave that transports the cortical output and parses it into frames, which produces the spatial texture of AM patterns.

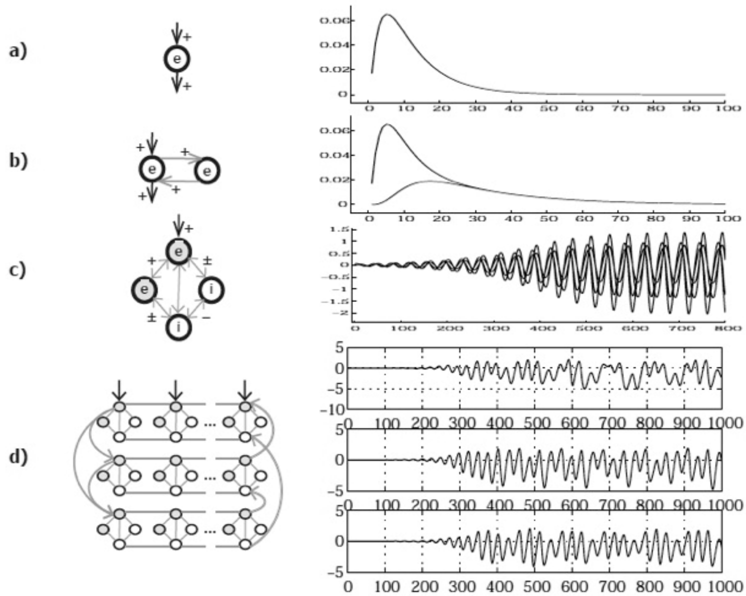

Figure 6. Illustration of the architecture and dynamics of Freeman K-sets; a) K0 set of non-interactive population with stable zero attractor; b) KI set with mutually excitatory neurons producing non-zero activation level; c) KII with interacting excitatory and inhibitory KI sets producing narrow-band oscillations; d) KIII set with interacting KII sets resulting in broad-band oscillations.

KIII sets are complex dynamical systems modelling brain dynamics in various sensory cortices. In their original forms, Freeman K-sets use a system of ordinary differential equations (ODEs). KIII is an example of associative memories with non-convergent dynamics. Dynamic KIII memories have several advantages as compared to convergent recurrent networks:

- They produce robust memories based on relatively few learning examples even in noisy environment;
- The encoding capacity of K-sets with a given number of nodes is exponentially larger than their convergent counterparts;
- K-sets can recall stored data very quickly, just as humans and animals can recognize a learned pattern within a fraction of a second.

Advanced K-sets include the KIV sets for modelling intentional behaviour. They are applicable to goal-oriented navigation and robot control. Finally, at the top level is the KV, which models the neocortex as a single organ as postulated by Williams James over a century ago. Freeman K-sets have been used successfully in describing the dynamic operation of sensory cortices and the formation of multisensory Gestalt percepts. The utility of K-sets has been demonstrated in a wide range of applications, including classification, time series prediction, and robot control and navigation.

Recent theoretical developments include rigorous modelling of the experimentally observed spatio-temporal synchronization–desynchronization transitions as large-scale coherence effects in the cortical sheet. Random Graph Theory (RGT) has been used to describe cortical phase transitions using a hierarchical model based on Freeman K-sets. RGT develops equations for the probability distributions of macroscopic state variables as an alternative to differential equations employed in the original implementation of K-sets. The modular neuropercolation model is motivated by the multilayer structure and dynamical properties of the cortex, and it describes critical brain oscillations corresponding to K0, KI, KII, and KIII sets, including background activity, narrow-band oscillations in excitatory-inhibitory populations, and broad-band oscillations in the cortex. Input-induced and spontaneous transitions between states with large-scale synchrony and without synchrony exhibit brief episodes with long-range spatial correlations as described in neuropercolation models, providing a plausible interpretation of the corresponding episodes in EEG and ECoG experiments with synchrony–desynchrony transitions as described in Freeman's cinematographic theory of perception. (See also the contribution by Kozma and Davis in this issue).

2.4. Singular Transitions in Brain Dynamics Modelled by Quantum Field Theory (QFT)

Brain functioning modelled by quantum fields

Freeman used the mathematical tools of classical physics for more than 40 years to model the functioning of brains, their 'dynamics', in small mammals by mapping the cortical electrical activity during intentional behaviour. At the level of classical physics, electric, magnetic, and diffusion fields were considered, but to Freeman they appeared to be insufficient for explaining the phenomena he was observing. Although classical nonlinear dynamics and statistical mechanics have produced great advances in brain studies, they are still lacking an in-depth explanation of brain functioning at the level of patterns made of a large number of neurons organized in synchronous oscillations modulated in phase and in amplitude, so-called AM patterns (in-phase or coherent oscillation patterns, in the jargon of quantum field theory — QFT).

There are indeed remarkably close correspondences between QFT and features and concepts in the observed brain functioning. QFT appears indeed to offer the mathematical tool that can explain the process of the formation, the *dynamic origin*, of long-range neuron correlations in AM patterns, their rapid formation and dissolution (*criticality*), their stability states compatible with such critical processes, different degrees of ordering in the AM patterns, and a convincing interpretation of the extremely high density energy consumption observed during memory retrieval in brains.

In 1967 Ricciardi and Umezawa proposed that the patterns observed in the brain originate from a mechanism called the 'spontaneous breakdown of symmetry' (SBS), by which ordered patterns are created in QFT. In 1995 the Ricciardi and Umezawa model was extended (Vitiello, 1995) to describe brain engagement with the external world (with *dissipative* dynamics). Brains indeed act into their environment to confirm or reject hypotheses, testing them by intentional action, as in trial-and-error learning. Their QFT modelling also includes the environment, so that every source and sink of energy for the brain is matched by a sink or source for the environment. This is achieved by doubling the variables that describe brain activity. The environment appears then to be a 'copy' of the system, its *Double*, in the sense that it receives/ gives what the system gives/receives. Such an exchange of roles is described by reversing the sign of time (time-reversal), since 'giving' for one of the systems (e.g. the brain) becomes, under time-reversal, 'receiving' for the other system (the environment), and vice versa.

Time-reversal dynamics reveal themselves in the observation that each AM pattern is accompanied by a pattern in the form of a cone exhibiting outward (exploding) or inward (imploding) time-reversed pulsations correlated in phase. The exploding gradient may be explained in terms of a pacemaker in conventional neurodynamics, but there is no classical explanation for the imploding gradient. The QFT model predicts the imploding regime and its exploding, time-reversed copy, in agreement with observations. In the last paper by Freeman, coauthored with one of us (Freeman and Vitiello, 2016), it was proposed that the AM patterns in forward time implement action on the environment, while their time-reversed copy governs perception, resulting in the mental awareness of 'to-be-in-the-world'. Forward in time and time-reversed activities are fully entangled in the action–perception cycle.

In the quantum dissipative model the neurons and other structures at the cellular level are considered to be classical objects, not quantum systems. The quantum degrees of freedom are the vibrational modes of the electrical dipoles characterizing bio-macromolecules and the water molecules, which are in number more than 90% of all molecular components in brains. SBS provides the bridge, or change of scale, from quantum dynamics to the behaviour of classical cells and their constructs. Brain studies using traditional classical tools are not superseded, but might be enhanced by understanding the underlying quantum dynamics. (This is further discussed in Vitiello's contribution in this issue.)

2.5. Broader Issues in Systems Theory, Consciousness, and the Unity of Mind-Brain-Body

In the pursuit of understanding the brain and its functions, it is inevitable that one needs to consider the relation between different organizational levels, in particular between neural and mental processes. However, since most of these relations are still unknown, the concepts used are often philosophical in character. Freeman has throughout the years produced highly influential results covering philosophical aspects of brain functions, including meaning, intentionality, consciousness, and free will, but also more generally on causality and the unity of mind and matter.

By pursuing to its roots the concept of intentionality, Freeman was led, via the philosophers Merleau-Ponty, Heidegger, and Dreyfus, to the medieval philosopher St. Thomas Aquinas, who was the first to introduce the term. Freeman notes that Aquinas revised Aristotelian deterministic philosophy in a way that emphasizes the role of individual choice and responsibility. In Aquinas's doctrine of intentionality, the self thrusts the body into the world, in search of stimuli, and by this the self changes itself to conform to the impact of the world. This process of learning he called 'assimilation', which begins with an act of exploration, followed by an internal change. Freeman found many similarities between these ideas and his own research, where the limbic system appeared to be the origin of the exploratory act, and where the patterns and neural structures are constantly changing as a result, even for the same stimulus.

Freeman pointed out that intentionality couldn't be explained by the common (since Aristotle) understanding of linear causality. Instead, *circular causality* could better explain it, in terms of action–perception cycles, in which each perception concomitantly is the outcome of a preceding action and the condition for a following action. Three properties of intentionality were identified — directedness, unity, and wholeness — which could in turn be associated with three stages of intentional behaviour:

1) In the first stage, a stimulus is categorized in the primary sensory cortex, usually accompanied by an experience of emotion and value;
2) In the second stage, different sensory modalities are integrated into *Gestalts*, which are assigned time and place in the life history of the individual;
3) In the third stage, there is a global coherence over the entire cerebral cortex, resulting in a decision to act (or not to act).

Intention is inevitable in evolution, and precedes consciousness, according to Freeman, but what is consciousness? Freeman never claimed he knew what it is, and he even considered consciousness a non-scientific issue, because you cannot measure it or prove that it exists in any other individual. Yet, in his 1991 *Scientific American* article, Freeman hypothesized that: 'Consciousness may well be the subjective experience of this recursive process of motor command, reafference and perception. If so, it enables the brain to plan and prepare for each subsequent action on the basis of past action, sensory input and perceptual synthesis.'

Later, Freeman specified his view on consciousness: 'Consciousness in the neurodynamic view is a global internal state variable composed of a sequence of momentary states of awareness… In humans, the operations and informational contents of the global state variable, which are sensations, images, feelings, thoughts and beliefs, constitute the experience of causation, which is to know intentionality in operation… (but) intentional acts do not require awareness, whereas voluntary acts require self-awareness' (Freeman, 1999).

Voluntary acts may be considered acts of free will, which, according to Freeman, is inherent in the phase transition, as a bifurcation, a branch point between two directions when making a decision, or a choice. It is inevitable, we cannot avoid making choices, from the very nature of the neurodynamics which must be resolved. No decision is ever totally free or determined, but it depends on you, as a subject. This is an expression of the operation of free will, which comes directly from Aquinas.

Free will can be seen as an extension of intentionality, when we become aware of our choices and hence responsible for our actions. In contrast to the dominant view in neuroscience today, Freeman strongly believed we are not determined by genes or the environment, because we have this additional property of individual choice. We are neither determined by natural laws nor under the complete influence of random events, but rather have a genuine capacity to make decisions, to make more or less rational choices out of our own free will. Even though the concept of free will is at least as complicated as the concept of consciousness, and the degrees of freedom from external influences may vary, Freeman believed the capacity of making choices is inherent in all mammals, if not all animals. (See also Liljenström's contribution in this issue.)

In recent years, Freeman made efforts to emphasize and show how the work of Aquinas on intentionality, meaning, mind, and action in the world should have strong and important implications for modern neuroscience. In a dialogue with the French neuroscientist Jean-Pierre Changeux in 2015, Freeman argued that determinism has to be abandoned and neuroscience has to recognize intentionality as an essential feature of organisms, from insects to man. While the role of Aquinas's philosophy in modern science was debated, they both agreed that consciousness has to be considered at different levels, depending on the complexity of the organism, where social interaction is of paramount importance. The operation of human society strongly depends on the capacity to imagine what other persons are like, what they are thinking, Freeman concluded in this dialogue (Freeman and Changeux, 2015).

Walter Freeman's philosophical work is an important extension of his groundbreaking neuroscience studies, reflecting upon questions of broader relevance to humanity. He beautifully summarizes the philosophical implications of his experimental and theoretical research:

> On the one hand, the main contribution of mesoscopic neurodynamics is to the self-organising chaotic patterns of gamma activity in primary sensory areas and the limbic system, which support the emergence of intentional behavior and make perception a creative act based in neural masses, not the passive information processing of neural networks. On the other hand, intentional behavior is largely unconscious, and where consciousness does supervene, it is a global macroscopic operator, which knits together the fabric of life history and supports the emergence of sophisticated value systems. (Freeman, 2005)

3. Further Developments

Von Neumann (1958), one of the inventors of digital computers, posed the following question about the computer and the brain: What is the language of the brain? In his answer, while appreciating the enormous potential of computers, he warned about a mechanistic parallel between brains and computers. He pointed out that the operation of brains follows principles other than the logical sequence of computational algorithms that have been used for the design of digital computers. Von Neumann contended that the language of the brain cannot be mathematics, as we know it. Freeman was fascinated by this question and deeply agreed with the argument against mechanistic parallels between the cold logic of digital computers and the complexity of our brains.

Following von Neumann's question on the language of the brain, Freeman's vision opens up new practical avenues for the design of artificially intelligent systems. In the robotics domain, the intentional action–perception cycle has been implemented in a novel control and navigation system for autonomous vehicles. Another domain of application is the development of novel brain–computer interface technologies, using the results on neural correlates of consciousness. Recent progress with advanced brain monitoring and modelling techniques provides a direct avenue to apply Freeman's results in the development of novel Brain Machine Interface technologies.

A great challenge to neuroscience is to describe and understand the relationship between neural and mental processes, between brain and consciousness. In this pursuit, it is essential to recognize different levels of consciousness, for example in relation to the complexity of the neural systems involved. Further studies of different animal species and their neurodynamics, as well as their behaviours, could give insight into this issue. Freeman, like many others, considered the evolution of the neocortex from the paleocortex a ‘quantum leap’ for consciousness. Another leap may be that between the brains of humans and other primates. The fairly recent findings that the human brain uniquely has neocortical areas which do not have connections to the basal ganglia (striatum and thalamus) could suggest the emergence of certain human characteristics. Freeman speculated that such a separation of brain structures, detached from sensory input to a degree not found in other animals, could allow for abstraction, for symbolic thinking, and for self-consciousness, and he termed these areas ‘koniocortex’ (from Greek *konio*, dust), since they have amorphous distributions of nerve cell nuclei so numerous as to resemble grains of dust. Very little is as yet known of these areas, and Freeman suggested further studies could give important insights in human cognition and consciousness.

More of Freeman’s legacy is presented by the authors of this paper, as well as by others, in this special issue, which consists of a collection of altogether 13 additional contributions.

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